

CHAPTER – 3

Graphics Algorithms

In case of the two dimensional algorithm, we mostly deal with the line, circle, ellipse etc as they are the basic output primitives. In case of line drawing, it is possible to draw a line using the direct mathematical formula for line: $y = mx + c$, but the main drawbacks of this algorithm are

- It required floating point addition and multiplication at each step while evaluating new position of pixel to put. In case of the DDA algorithm we calculate 'm' in only one time.
- In each computational step, coordinate value of new pixel position required to round off which leads staircase effect.
- The line is drawn only interpreting the nature of equation but not analyze the case of $\Delta x > \Delta y$ or $\Delta x < \Delta y$ as DDA.

3.1 Digital Differential Analyzer (DDA) Algorithm

In the DDA algorithm, we have to approximate the value of each pixel in whole number if any value of x-coordinate or y-coordinate is in fractional form by the addition or subtraction of the slope (m). Here, we approximate the value by the round-off or by counting the ceiling or floor value. This is the main disadvantage of the DDA algorithm of deviate the line from its exact position.

Advantages of DDA algorithm

- DDA algorithm is faster method for calculating pixel position than simple line drawing algorithm
- It avoids directly deal of equation of straight line $y = mx + c$
- It avoids the floating point addition and subtraction in the simple algorithm

Limitations in DDA Algorithm

- The slope "m" is usually stored in floating point number
- There could be round off error
- The line will move away from the true line path, especially when it is long due to successive round of error.

3.1.1 Positive Slope

In case of the positive slope, the portion in x-direction & y-direction equally propagates.

Here,

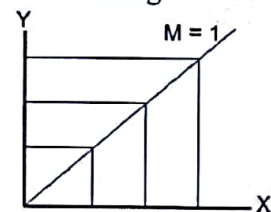


Fig. Positive Slope

1. Moving Left to Right

In moving left to right, we assume the leftmost point as the first and the rightmost point as the final point.

$$\Delta y = m \Delta x$$

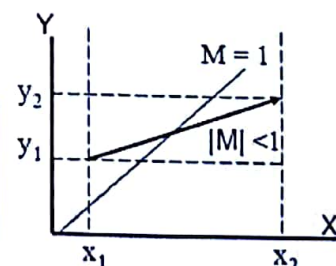
$$\Delta x = \frac{\Delta y}{m}$$

a. When slope (m) < 1

Here, the portion in the x-direction is relatively increase largely than that of the portion in y-direction.

Hence,

$\Delta x = 1$	$x_{k+1} = x_k + 1$
$\Delta y = m$	$y_{k+1} = y_k + m$



b. When slope (m) > 1

Here, the portion in the y-direction is relatively increase largely than that of the portion in x-direction.

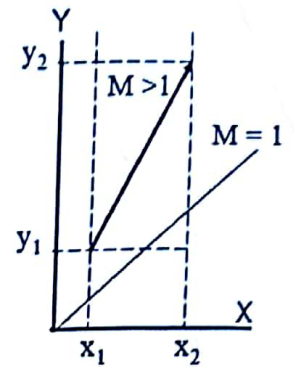
Hence,

$$\Delta x = \frac{1}{m}$$

$$\Delta y = 1$$

$$x_{k+1} = x_k + \frac{1}{m}$$

$$y_{k+1} = y_k + 1$$



2. Moving Right to Left

In moving right to left, we assume the rightmost point as the first and the leftmost point as the final point.

a. When slope (m) < 1

Here, the portion in the x-direction is relatively decreases largely than that of the portion in y-direction.

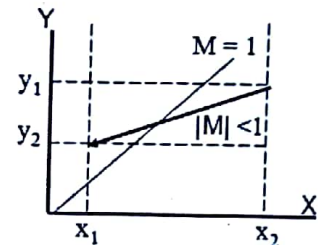
Hence,

$$\Delta x = -1$$

$$\Delta y = -m$$

$$x_{k-1} = x_k - 1$$

$$y_{k-1} = y_k - m$$



b. When slope (m) > 1

Here, the portion in the y-direction is relatively decreases largely than that of the portion in x-direction.

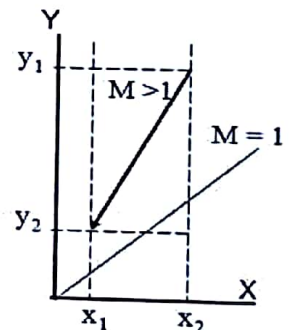
Hence,

$$\Delta x = -\frac{1}{m}$$

$$\Delta y = -1$$

$$x_{k-1} = x_k - \frac{1}{m}$$

$$y_{k-1} = y_k - 1$$



3.1.2 Negative Slope

In case of the negative slope, the portion in x-direction & y-direction are not equally propagates as shown in figure.

Here,

$$\Delta y = m \Delta x$$

$$\Delta x = \frac{\Delta y}{m}$$

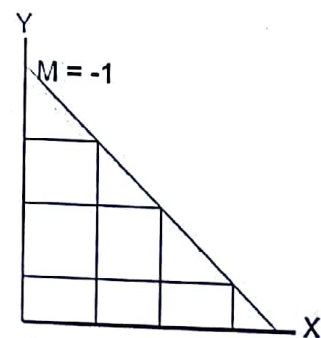


Fig. Negative Slope

3. Moving Left to Right

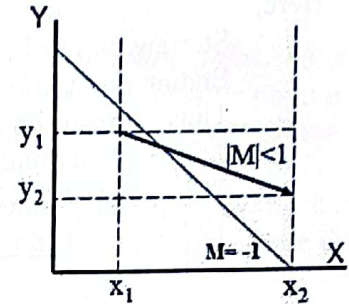
In moving left to right, we assume the leftmost point as the first and the rightmost point as the final point.

a. When slope (m) < 1

Here, the portion in the x-direction is relatively increase largely than that of the portion in y-direction.

Hence,

$$\begin{aligned} \Delta x &= 1 \\ \Delta y &= -m \end{aligned} \quad \begin{aligned} x_{k+1} &= x_k + 1 \\ y_{k+1} &= y_k - m \end{aligned}$$

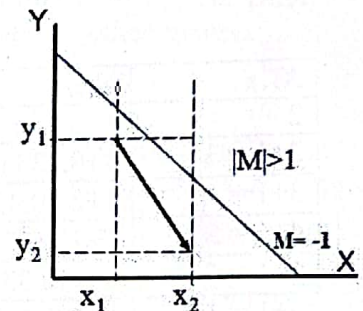


b. When slope (m) > 1

Here, the portion in the y-direction is relatively increase largely than that of the portion in x-direction.

Hence,

$$\begin{aligned} \Delta x &= \frac{1}{m} \\ \Delta y &= -1 \end{aligned} \quad \begin{aligned} x_{k+1} &= x_k + \frac{1}{m} \\ y_{k+1} &= y_k - 1 \end{aligned}$$



4. Moving Right to Left

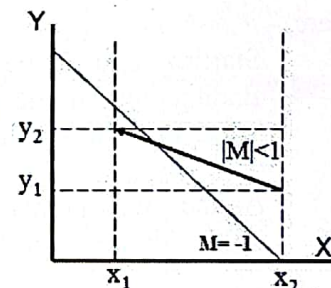
In moving right to left, we assume the rightmost point as the first and the leftmost point as the final point.

a. When slope (m) < 1

Here, the portion in the x-direction is relatively decreases largely than that of the portion in y-direction.

Hence,

$$\begin{aligned} \Delta x &= -1 \\ \Delta y &= m \end{aligned} \quad \begin{aligned} x_{k+1} &= x_k - 1 \\ y_{k+1} &= y_k + m \end{aligned}$$

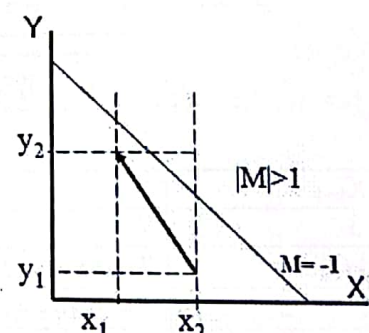


b. When slope (m) > 1

Here, the portion in the y-direction is relatively decreases largely than that of the portion in x-direction.

Hence,

$$\begin{aligned} \Delta x &= -\frac{1}{m} \\ \Delta y &= 1 \end{aligned} \quad \begin{aligned} x_{k+1} &= x_k - \frac{1}{m} \\ y_{k+1} &= y_k + 1 \end{aligned}$$



Note: The DDA algorithm is a faster method for calculating the pixel position than that the use of direct method of line drawing method by using $y = mx + c$. The DDA method is seems good as it do not need to calculate the slope (m), in each iteration like the direct method.

Example-1: Digitize the line with end points (2, 1) and (8, 3) using DDA.

Solution

Here,

Starting point of line = $(x_1, y_1) = (2, 1)$ and

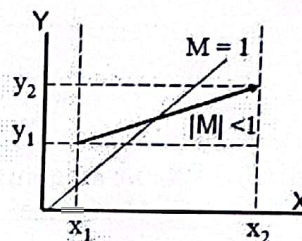
Ending point of line = $(x_2, y_2) = (8, 3)$

Thus, slope of line, $m = \Delta y / \Delta x = y_2 - y_1 / x_2 - x_1 = (3-1) / (8-2) = 1/3 = 0.3333$

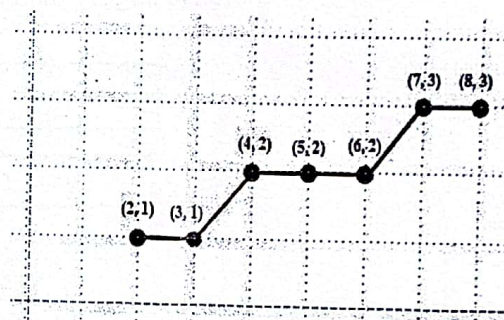
As the given points, it is clear that the line is moving left to right with the slope $m = 1/3 < 1$

Thus,

$$\begin{array}{|c|} \hline \Delta x = 1 \\ \hline \Delta y = m \\ \hline \end{array} \quad \begin{array}{|c|} \hline X_{k+1} = X_k + 1 \\ \hline Y_{k+1} = Y_k + m \\ \hline \end{array}$$



X_{k+1}	Y_{k+1}	(X_{k+1}, Y_{k+1})
2	1	(2, 1)
2+1 = 3	$1 + 0.3333 = 1.333 \approx 1$	(3, 1)
3+1 = 4	$1 + 2 * 0.3333 = 1.666 \approx 2$	(4, 2)
4+1 = 5	$1 + 3 * 0.3333 = 1.999 \approx 2$	(5, 2)
5+1 = 6	$1 + 4 * 0.3333 = 2.333 \approx 2$	(6, 2)
6+1 = 7	$1 + 5 * 0.3333 = 2.666 \approx 3$	(7, 3)
7+1 = 8	$1 + 6 * 0.3333 = 2.999 \approx 3$	(8, 3)



Example-2: Digitize the line with end points (3, 7) and (8, 3) using DDA.

Solution

Here,

Starting point of line = $(x_1, y_1) = (3, 7)$ and

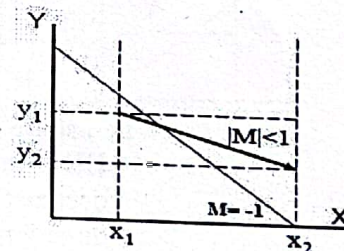
Ending point of line = $(x_2, y_2) = (8, 3)$

Thus, slope of line, $m = \Delta y / \Delta x = y_2 - y_1 / x_2 - x_1 = (3-7) / (8-3) = -4/5 = -0.8$

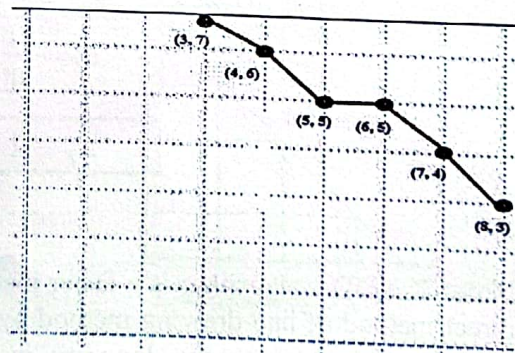
As the given points, it is clear that the line is moving left to right with the negative slope, $|m| = 0.8 < 1$

Thus,

$$\begin{array}{|c|} \hline \Delta x = 1 \\ \hline \Delta y = -m \\ \hline \end{array} \quad \begin{array}{|c|} \hline X_{k+1} = X_k + 1 \\ \hline Y_{k+1} = Y_k - m \\ \hline \end{array}$$



X_{k+1}	Y_{k+1}	(X_{k+1}, Y_{k+1})
3	7	(3, 7)
3+1 = 4	$7 - 0.8 = 6.2 \approx 6$	(4, 6)
4+1 = 5	$7 - 2 * 0.8 = 5.4 \approx 5$	(5, 5)
5+1 = 6	$7 - 3 * 0.8 = 4.6 \approx 5$	(6, 5)
6+1 = 7	$7 - 4 * 0.8 = 3.8 \approx 4$	(7, 4)
7+1 = 8	$7 - 5 * 0.8 = 3$	(8, 3)



3.2 Bresenham's Line Algorithm (BLA)

The BLA is a more efficient method used to plot pixel position along a straight-line path.

3.3.1 Advantage of BLA over DDA

- In DDA algorithm each successive point is computed in *floating point*, so it requires *more time* and *more memory space*. While in BLA each successive point is calculated in *integer value* or *whole number*. So it requires less time and less memory space.
- In DDA, since the calculated point value is floating point number, it should be rounded at the end of calculation but in BLA it does not need to round, so there is no accumulation of rounding error.
- Due to rounding error, the line drawn by DDA algorithm is not accurate, while in BLA line is accurate.
- DDA algorithm cannot be used in other application except line drawing, but BLA can be implemented in other application such as circle, ellipse and other curves.

3.2.2 Bresenham's Line drawing Algorithm for slope, $|m| \leq 1$

Let us assume that pixel (x_k, y_k) is already plotted assuming that the *sampling direction* is along X-axis i.e. (x_k+1, y_k) or (x_k+1, y_k+1) . Thus, the common equation of the line is $y = m(x_k+1) + c$.

Now,

$$d_1 = y - y_k = m(x_k+1) + c - y_k \text{ and}$$

$$d_2 = y_{k+1} - y = y_k+1 - \{m(x_k+1) + c\}$$

$$\text{So, } d_1 - d_2 = [m(x_k+1) + c - y_k] - [y_k+1 - \{m(x_k+1) + c\}]$$

$$\text{Or, } d_1 - d_2 = 2m(x_k+1) + 2c - 2y_k - 1$$

Since, slope of line $(m) = \Delta y / \Delta x$, we have

$$P_k = \Delta x (d_1 - d_2) = 2\Delta y (x_k+1) + 2c \Delta x - 2\Delta x y_k - \Delta x$$

$$P_k = 2\Delta y x_k - 2\Delta x y_k + \{2\Delta y + 2c \Delta x - \Delta x\}$$

$$P_k = 2\Delta y x_k - 2\Delta x y_k + b, \text{ equation - I and } (K+1)^{\text{th}} \text{ step, we have}$$

$$P_{k+1} = 2\Delta y x_{k+1} - 2\Delta x y_{k+1} + b, \text{ equation - II}$$

Subtracting equation I & II, we have

$$P_{k+1} - P_k = 2\Delta y (x_{k+1} - x_k) - 2\Delta x (y_{k+1} - y_k)$$

As we are sampling in x-direction, increase by 1-pixel position in left to right i.e. $x_{k+1} = x_k+1$.

We have,

$$P_{k+1} = P_k + 2\Delta y (x_{k+1} - x_k) - 2\Delta x (y_{k+1} - y_k)$$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x (y_{k+1} - y_k), \text{ equation - III}$$

Now,

- If $P_k > 0$, (i.e. $d_1 - d_2 = \text{Positive}$ or $y_{k+1} - y_k = 1$), now equation - III reduces as:

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x, \text{ and } y_{k+1} = y_k + 1 \text{ \& } x_{k+1} = x_k + 1$$

- Else

If $P_k < 0$, (i.e. $d_1 - d_2 = \text{Negative}$ or $y_{k+1} - y_k = 0$), now equation - III reduces as:

$$P_{k+1} = P_k + 2\Delta y, \text{ and } y_{k+1} = y_k \text{ \& } x_{k+1} = x_k + 1$$

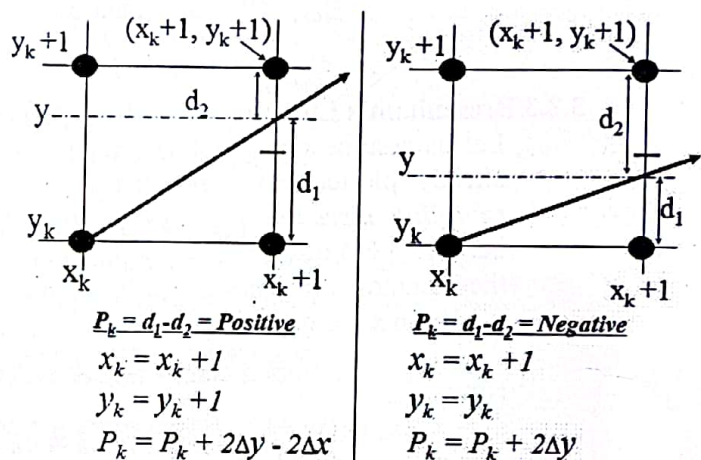


Fig. Bresenham's Line Algorithm for $|m| \leq 1$

Note: If the starting point is (x_0, y_0) , then what will be the value of initial decision parameter (P_0) when $|m| \leq 1$?

We have $P_k = \Delta x (d_1 - d_2)$

$$P_k = \Delta x (2m(x_k+1) + 2c - 2y_k - 1) \text{ [since, } d_1 - d_2 = 2m(x_k+1) + 2c - 2y_k - 1]$$

$$P_k = \Delta x (2mx_k + 2m + 2c - 2y_k - 1)$$

$$P_k = \Delta x \{2(mx_k + c - y_k) + 2m - 1\}$$

The line $y = mx + c$ must pass through the initial point (x_0, y_0) and hence $(mx_k + c - y_k) = 0$, we have

$$\text{Thus, } P_0 = \Delta x (2m - 1)$$

$$P_0 = \Delta x (2\Delta y / \Delta x - 1) \text{ [since, } m = \Delta y / \Delta x]$$

$$P_0 = 2\Delta y - \Delta x, \text{ which is the initial decision parameter.}$$

Conclusion: Bresenham's Line drawing Algorithm for slope, $|m| \leq 1$

- Calculate the starting value of the decision parameter: $P_0 = 2\Delta y - \Delta x$.
- At each x_k along the line, starting at $k = 0$, perform the following test.

- If $P_k > 0$,

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x, \text{ and } y_{k+1} = y_k + 1 \text{ \& } x_{k+1} = x_k + 1$$

- Else If $P_k < 0$,

$$P_{k+1} = P_k + 2\Delta y, \text{ and } y_{k+1} = y_k \text{ \& } x_{k+1} = x_k + 1$$

3.2.3 Bresenham's Line drawing Algorithm for slope, $|m| > 1$

Let us assume that pixel (x_k, y_k) is already plotted assuming that the **sampling direction** is along y-axis i.e. (x_k, y_k+1) or (x_k+1, y_k+1) . Thus, the common equation of the line is $y_k+1 = mx + c$ or $x = \{y_k+1 - c\}/m$

Now,

$$d_1 = x - x_k = \{y_k+1 - c\}/m - x_k \text{ and } d_2 = x_{k+1} - x = x_{k+1} - \{y_k+1 - c\}/m$$

So,

$$d_1 - d_2 = [\{y_k+1 - c\}/m - x_k] - [x_{k+1} - \{y_k+1 - c\}/m]$$

$$\text{Or, } d_1 - d_2 = 2/m * (y_k+1) - 2c/m - 2x_k - 1$$

Since, slope of line $(m) = \Delta y / \Delta x$, we have

$$P_k = \Delta y (d_1 - d_2) = 2\Delta x (y_k+1) - 2c\Delta x - 2\Delta y x_k - \Delta y$$

$$P_k = 2\Delta x y_k - 2\Delta y x_k + \{2\Delta x - 2c\Delta x - \Delta y\}$$

$$P_k = 2\Delta x y_k - 2\Delta y x_k + b, \text{ equation - I and } (K+1)^{\text{th}} \text{ step, we have}$$

$$P_{k+1} = 2\Delta x y_{k+1} - 2\Delta y x_{k+1} + b, \text{ equation - II}$$

Subtracting equation I & II, we have

$$P_{k+1} - P_k = 2\Delta x (y_{k+1} - y_k) - 2\Delta y (x_{k+1} - x_k)$$

As we are sampling in y-direction, increase by 1-pixel position in lower to upper position i.e. $y_{k+1} = y_k + 1$.

We have,

$$P_{k+1} = P_k + 2\Delta x (y_k+1 - y_k) - 2\Delta y (x_{k+1} - x_k)$$

$$P_{k+1} = P_k + 2\Delta x - 2\Delta y (x_{k+1} - x_k), \text{ equation - III}$$

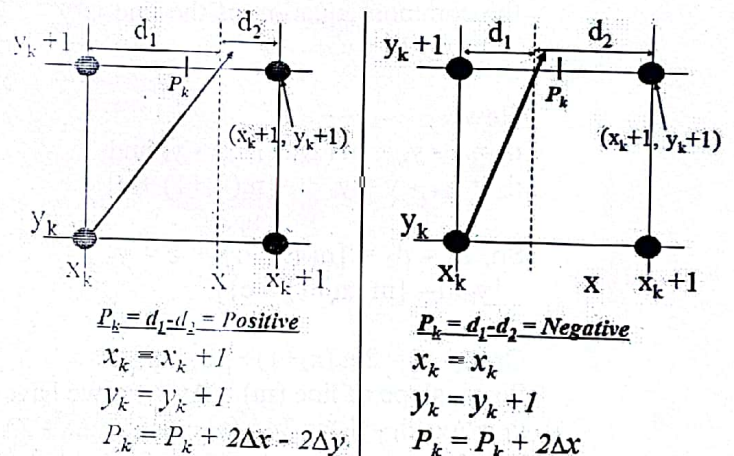


Fig. Bresenham's Line Algorithm for $|m| > 1$

Now,

- If $P_k > 0$, (i.e. $d_1 - d_2 = \text{Positive}$ or $x_{k+1} - x_k = 1$), now equation - III reduces as:
 $P_{k+1} = P_k + 2\Delta x - 2\Delta y$, and $x_{k+1} = x_k + 1$ & $y_{k+1} = y_k + 1$
- Else

If $P_k < 0$, (i.e. $d_1 - d_2 = \text{Negative}$ or $x_{k+1} - x_k = 0$), now equation - III reduces as:

$$P_{k+1} = P_k + 2\Delta x, \text{ and } x_{k+1} = x_k \text{ \& } y_{k+1} = y_k + 1$$

Note: If the starting point is (x_0, y_0) , then what will be the value of initial decision parameter (P_0) when $|m| > 1$?

We have

$$P_k = \Delta y (d_1 - d_2)$$

$$P_k = \Delta y \{2/m * (y_k + 1) - 2c/m - 2x_k - 1\} \text{ [since, } d_1 - d_2 = 2/m * (y_k + 1) - 2c/m - 2x_k - 1]$$

$$P_k = \Delta y \{2y_k/m + 2/m - 2c/m - 2x_k - 1\}$$

$$P_k = \Delta y [2/m \{y_k - c - mx_k\} + 2/m - 1]$$

The line $y = mx + c$ must pass through the initial point (x_0, y_0) and hence $y_0 - c - mx_0 = 0$, we have

Thus,

$$P_0 = \Delta y (2/m - 1)$$

$$P_0 = \Delta y (2\Delta x / \Delta y - 1) \text{ [since, } m = \Delta y / \Delta x]$$

$$P_0 = 2\Delta x - \Delta y, \text{ which is the initial decision parameter.}$$

Conclusion: Bresenham's Line drawing Algorithm for slope, $|m| > 1$

- Calculate the starting value of the decision parameter: $P_0 = 2\Delta y - \Delta x$.
- At each y_k along the line, starting at $k = 0$, perform the following test.

- If $P_k > 0$,
 $P_{k+1} = P_k + 2\Delta x - 2\Delta y$, and $x_{k+1} = x_k + 1$ & $y_{k+1} = y_k + 1$
- Else If $P_k < 0$,
 $P_{k+1} = P_k + 2\Delta x$, and $x_{k+1} = x_k$ & $y_{k+1} = y_k + 1$

Example-2: Digitize the line with end points (20, 10) and (30, 18) using BLA.

Solution

Here, Starting point of line = $(x_1, y_1) = (20, 10)$ and

Ending point of line = $(x_2, y_2) = (30, 18)$

Thus, slope of line, $m = \Delta y / \Delta x = y_2 - y_1 / x_2 - x_1 = (18 - 10) / (30 - 20) = 8/10$

As the given points, it is clear that the line is moving left to right with the negative slope,
 $|m| = 0.8 < 1$

Thus,

The initial decision parameter (P_0) = $2\Delta y - \Delta x = 2*8 - 10 = 6$

Since, for the Bresenham's Line drawing Algorithm of slope, $|m| \leq 1$, we have

- If $P_k > 0$,
 $P_{k+1} = P_k + 2\Delta y - 2\Delta x$, and $y_{k+1} = y_k + 1$ & $x_{k+1} = x_k + 1$
- Else
 If $P_k < 0$,
 $P_{k+1} = P_k + 2\Delta y$, and $y_{k+1} = y_k$ & $x_{k+1} = x_k + 1$

Thus,

k	P_k	X_{k+1}	Y_{k+1}	(X_{k+1}, Y_{k+1})
0.	6	$20+1 = 21$	11	(21, 11)
1.	$6 + 2*8 - 2*10 = 2$	$21+1 = 22$	11	(22, 11)
2.	$2 + 2*8 - 2*10 = -2$	$22+1 = 23$	12	(23, 12)
3.	$-2 + 2*8 = 14$	$23+1 = 24$	13	(24, 13)
4.	$14 + 2*8 - 2*10 = 10$	$24+1 = 25$	14	(25, 14)
5.	$10 + 2*8 - 2*10 = 6$	$25+1 = 26$	15	(26, 15)
6.	$6 + 2*8 - 2*10 = 2$	$26+1 = 27$	15	(27, 15)
7.	$2 + 2*8 - 2*10 = -2$	$27+1 = 28$	16	(28, 16)
8.	$-2 + 2*8 = 14$	$28+1 = 29$	17	(29, 17)
9.	$14 + 2*8 - 2*10 = 10$	$29+1 = 30$	18	(30, 18)

Example-2: Digitize the line with end points (1, 0) and (3, 3) using BLA.

Solution

Here, Starting point of line = $(x_1, y_1) = (1, 0)$ and

Ending point of line = $(x_2, y_2) = (3, 3)$

Thus, slope of line, $m = \Delta y / \Delta x = y_2 - y_1 / x_2 - x_1 = (3-0) / (3-1) = 3/2$

As the given points, it is clear that the line is moving left to right with the negative slope, $|m| = 1.5 > 1$

Thus, the initial decision parameter $(P_0) = 2\Delta x - \Delta y = 2*2 - 3 = 1$

Since, for the Bresenham's Line drawing Algorithm of slope, $|m| > 1$, we have

- If $P_k > 0$,
 $P_{k+1} = P_k + 2\Delta x - 2\Delta y$, and $x_{k+1} = x_k + 1$ & $y_{k+1} = y_k + 1$
- Else If $P_k \leq 0$,
 $P_{k+1} = P_k + 2\Delta x$, and $x_{k+1} = x_k$ & $y_{k+1} = y_k + 1$

Thus,

k	P_k	X_{k+1}	Y_{k+1}	(X_{k+1}, Y_{k+1})
0.	1	2	1	(2, 1)
1.	$1 + 2*2 - 2*3 = -1$	2	$1+1 = 2$	(2, 2)
2.	$-1 + 2*2 = 3$	3	$2+1 = 3$	(3, 3)

3.3 Midpoint Circle Algorithm:

- Let us consider a circle centered at origin. Assume that (x, y) is any point on the circle we can trivially compute other seven points on the circle as shown in figure.
- The slope of the circular curve is less than zero inside the circle, equal to zero on the circle boundary and greater than zero on outside the circle.
- We can generate all pixel positions around a circle by calculating only the points within the first octants & positions in the other seven octants are obtained by symmetry.
- Here, we have to calculate all the pixel position from $(0, r)$ up to both the co-ordinate either equal or x-coordinate is greater than that of the y-coordinate but the final value cannot be counted in the octant coordinate (i.e. only for checking) because this point is the starting pixel for next octant.

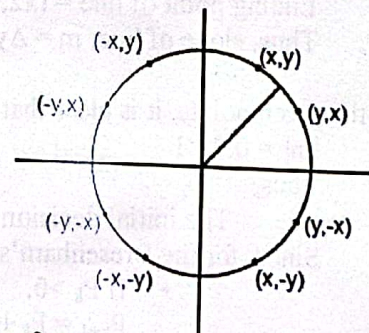


Fig. 8 symmetrical points on circle

To apply the midpoint method, we define a circle function:

$$f_{\text{circle}}(x, y) = x^2 + y^2 - r^2$$

The relative position of any point (x, y) can be determined by checking the sign of the circle function as: -

- $f_{\text{circle}}(x, y) < 0$, the point (x, y) is inside the circle boundary.
- $f_{\text{circle}}(x, y) = 0$, the point (x, y) is on the circle boundary.
- $f_{\text{circle}}(x, y) > 0$, the point (x, y) is outside the circle boundary.

Suppose, (x_k, y_k) is the pixel plotted, then the next pixel (x_{k+1}, y_{k+1}) will be either (x_{k+1}, y_k) or (x_{k+1}, y_{k-1}) as shown in figure.

Here, the mid-point $(m) = (x_k+1, y_k-1/2)$.

Thus, our decision parameter $(P_k) = f_{\text{circle}}(x_k+1, y_k-1/2) = (x_k+1)^2 + (y_k-1/2)^2 - r^2 \rightarrow I$, is the circle function evaluated at midpoint.

Also, the successive decision parameters are obtained using incremental calculations. We obtain a recursive expression for the next decision parameter by evaluation the circle function at sampling position $x_{k+1}+1 = x_k+2$. Thus, the next pixel to plot will either be $(x_{k+1}+1, y_k)$ or $(x_{k+1}+1, y_{k-1})$.

Also, the midpoint here is $(x_{k+1}+1, y_{k+1}-1/2)$ and thus, our decision parameter $(P_{k+1}) = f_{\text{circle}}(x_{k+1}+1, y_{k+1}-1/2)$

$$P_{k+1} = (x_{k+1}+1)^2 + (y_{k+1}-1/2)^2 - r^2$$

$$P_{k+1} = \{(x_k+1)+1\}^2 + (y_{k+1}-1/2)^2 - r^2$$

$$P_{k+1} = (x_k+1)^2 + 2(x_k+1) + 1 + (y_{k+1}-1/2)^2 - r^2 \rightarrow II$$

Now, subtracting I from II, we have

$$P_{k+1} - P_k = 2(x_k+1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

If $P_k < 0$, the midpoint is inside the circle and the pixel on the scan line y_k closer to the circle boundary. Otherwise, the midpoint is outside or on the circle boundary, and we select the pixel on scan line y_{k-1} .

Case - I: $P_k < 0$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k$$

$$P_{k+1} = P_k + 2(x_k+1) + 1$$

$$P_{k+1} = P_k + 2x_{k+1} + 1$$

Case - I: $P_k \geq 0$

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k - 1$$

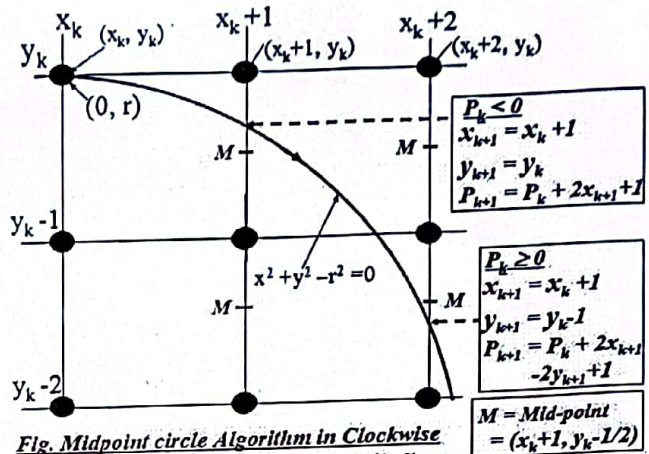
$$P_{k+1} = P_k + 2(x_k+1) + [(y_k-1)^2 - y_k^2] - (y_k-1 - y_k) + 1$$

$$= P_k + 2(x_k+1) + [(y_k^2 - 2y_k + 1 - y_k^2)] - (y_k-1 - y_k) + 1$$

$$= P_k + 2(x_k+1) - 2y_k + 1 + 1 + 1$$

$$= P_k + 2(x_k+1) - 2(y_k+1) + 1$$

$$P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1$$



3.3.1 Initial decision parameter (P_0) for circle

The initial decision parameter is obtained by evaluating the circle function at the start position $(x_0, y_0) = (0, r)$. Here, the next pixel will either be $(1, r)$ or $(1, r-1)$ where the midpoint is $(1, r-1/2)$. Thus, the initial decision parameter is given by:

$$P_0 = f_{\text{circle}}(1, r-1/2) = 1^2 + (r-1/2)^2 - r^2 = 5/4 - r \approx 1 - r$$

Thus, $P_0 = 1 - r$

Conclusion

1. Calculate the initial decision parameter $(P_0) = 1 - r$
2. If $P < 0$

Plot $(x_k + 1, y_k)$

$$P_{k+1} = P_k + 2x_{k+1} + 1$$

Else ($P \geq 0$)

Plot $(x_k + 1, y_k - 1)$

$$P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1$$

Note : If the center of the circle is not at origin then first calculate the octant points of the circle in the same way as the center at origin & then add the given circle center on each calculated pixels.

Example: Digitize a circle with radius 10.

Solution

Here, the initial decision parameter $(P_0) = 1 - r = 1 - 10 = -9$

Since, for the Midpoint Circle Algorithm of initial point $(0, r)$ & center at origin $(0, 0)$ rotating at clockwise direction, we have

If $P < 0$

Plot $(x_k + 1, y_k)$

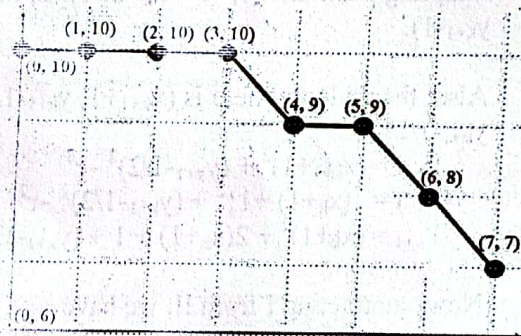
$$P_{k+1} = P_k + 2x_{k+1} + 1$$

Else ($P \geq 0$)

Plot $(x_k + 1, y_k - 1)$

$$P_{k+1} = P_k + 2x_{k+1} - 2y_{k+1} + 1$$

Thus,



k	P_k	x_{k+1}	y_{k+1}	(x_{k+1}, y_{k+1})
0.	-9	$0+1=1$	10	(1, 10)
1.	$-9+2*1+1=-6$	$1+1=2$	10	(2, 10)
2.	$-6+2*2+1=-1$	$2+1=3$	10	(3, 10)
3.	$-1+2*3+1=6$	$3+1=4$	9	(4, 9)
4.	$6+2*4-2*9+1=-3$	$4+1=5$	9	(5, 9)
5.	$-3+2*5-2*8+1=8$	$5+1=6$	8	(6, 8)
6.	$8+2*6-2*8+1=5$	$6+1=7$	7	(7, 7) (Halt)